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Bihar Engineering University, Patna

B.Tech 1st Semester Examination, 2024

Time: 03 Hours

Code: 100101 ***Subject: Engineering Chemistry*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option/answer the following (Any seven question only): **[14]**

(a) A sp hybrid orbital contains
- (i) 25% s-character
- (ii) 75% s-character
- (iii) 50% s-character
- (iv) 100% s-character

(b) Which one is least energetic radiation in Electromagnetic Spectrum.
- (i) Gamma rays
- (ii) X-Rays
- (iii) IR Rays
- (iv) Radio waves

(c) The C=*O* stretching vibration typically appears in the IR spectrum around:
- (i) 1600 cm^{-1}
- (ii) 3000 cm^{-1}
- (iii) 2200 cm^{-1}
- (iv) 3600 cm^{-1}

(d) What happens to the $lambda_{max}$ when conjugation in a molecule increases?
- (i) It shifts to lower wavelengths (blue shift)
- (ii) It shifts to higher wavelengths (red shift)
- (iii) Both of Them
- (iv) None of Them

(e) The standard hydrogen electrode (SHE) is assigned a potential of:
- (i) +1.0 V
- (ii) -1.0 V
- (iii) 0.00 V
- (iv) None of them

(f) Scale formation in boilers is due to:
- (i) Presence of dissolved gases like CO_2
- (ii) Presence of dissolved salts like $CaSO_4$
- (iii) Presence of suspended solids
- (iv) Presence of Bacteria

(g) Alkalinity in water is primarily due to the presence of
- (i) Chlorides and sulfates
- (ii) Calcium and magnesium
- (iii) Hydroxide, carbonate, and bicarbonate ions
- (iv) Nitrates and phosphates

(h) Which one is a example of thermosetting plastic?
- (i) Polyethylene
- (ii) PVC
- (iii) Bakelite
- (iv) Teflon

(i) Which of the following is a biodegradable polymer?
- (i) PVA
- (ii) PVC
- (iii) Teflon
- (iv) PHB (Polyhydroxybutyrate)

(j) The product of a Diels-Alder reaction is
- (i) Linear alkene
- (ii) Six-membered ring
- (iii) Four-membered ring
- (iv) Five-membered ring

Q.2 Solve both questions : **[14]**

(a) Discuss the nature of metal-carbon bonding (synergistic effect) in metal carbonyls.

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(b) What is VSEPR theory? Give its important postulates & predict the structure of H_2O, NH_3 & CH_4 using this theory.

Q.3 Solve both questions : **[14]**

(a) Write note on different types of electronic transition caused by absorption of energy in the UV region.

(b) Explain Chemical shift in NMR spectroscopy. Discuss various factors influencing the chemical shift.

Q.4 Solve both questions : **[14]**

(a) Discuss in brief about Chemical or Dry corrosion.

(b) What is proximate and ultimate analysis of water. Explain how proximate analysis of a coal sample is done?

Q.5 Solve both questions : **[14]**

(a) Discuss in brief about the Internal treatment methods for prevention of scale formation in boiler.

(b) What do you mean by Hardness of water. Calculate the temporary & permanent hardness in degree Clark of a water sample having the following analysis: $Mg(HCO_3)_2$= 146~mg/L, $Ca(HCO_3)_2$= 162~mg/L, $CaSO_4$= 136~mg/L, $MgCl_2$= 95~mg/L, $CaCl_2$= 111~mg/L & $NaCl$ = 50~mg/L.

Q.6 Solve both questions : **[14]**

(a) Calculate the amount of lime (92% pure) and soda (98% pure) required for treatment of 30000 litre of water whose analysis is as follows: $Mg(HCO_3)_2$= 36.5 ppm, $Ca(HCO_3)_2$= 40.5 ppm, $MgSO_4$= 30.0 ppm, $CaCl_2$= 27.75 ppm, $CaSO_4$= 34.0 & $NaCl$ = 10.0 ppm.

(b) Give properties & uses of the polyethylene, polyvinyl Chloride & Buna-S.

Q.7 Solve both questions : **[14]**

(a) How are polymerization reactions classified on the basis of mechanism. Discuss the mechanism of free radical polymerization reaction.

(b) Differentiate between Thermoplastic & thermosetting resins. Write the synthesis & application of phenol-formaldehyde resin.

Q.8 Solve both questions : **[14]**

(a) Discuss the mechanism of Diels Elder ring Cyclization reaction.

(b) Describe the mechanism of preparation of Aspirin.

Q.9 Write short notes on any two of the following: **[14]**

(a) Heisenberg's uncertainty principle

(b) Photoelectric effect

(c) Zeolite Process

(d) Silicons

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